

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I T.chaitra (192425217) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T.chaitra

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

AI-Assisted Medical Diagnosis using Google's AMIE (Articulate Medical Intelligence Explorer)

- **Domain:** Healthcare
- **AI Technologies:** Large Language Models (LLMs), Natural Language Processing (NLP), Machine Learning, Generative AI
- **Why this topic?** It is one of the latest AI healthcare applications (2024–2025), developed by Google Research, with many recent research papers and real-world relevance.

AI in Healthcare Report

1. Title of AI Application

Title: AI-Assisted Medical Diagnosis using Google's AMIE (Articulate Medical Intelligence Explorer)

AI Domain: Healthcare

2. Domain Selection

Application Domain: Healthcare

AMIE is an AI-powered conversational medical assistant designed to support clinical diagnosis by interacting with patients, collecting symptoms, suggesting possible diagnoses, and assisting healthcare professionals.

3. Introduction and Background

Artificial Intelligence is transforming healthcare by improving diagnostic accuracy, reducing workload for doctors, and providing faster medical assistance.

Google Research introduced **AMIE (Articulate Medical Intelligence Explorer)**, a conversational AI system capable of conducting diagnostic conversations similar to a physician. It uses advanced Large Language Models (LLMs) to understand patient symptoms, ask follow-up questions, and recommend possible diagnoses.

Its purpose is to improve healthcare accessibility, particularly in regions with limited medical professionals.

4. Problem Identification

Healthcare systems face several challenges:

- Shortage of doctors
- Long waiting times
- Delayed diagnosis
- Human errors
- Increasing number of patients
- Limited healthcare access in rural areas

Traditional diagnosis depends heavily on physician availability and experience.

5. Need for AI Solution

AI helps overcome these challenges by:

- Providing 24/7 medical assistance
- Improving diagnostic accuracy
- Supporting doctors in clinical decisions

- Reducing diagnostic errors
- Saving consultation time
- Assisting remote healthcare

Compared to traditional diagnosis, AI processes vast amounts of medical knowledge quickly and consistently.

6. AI Technologies Used

The AMIE system combines multiple AI technologies:

- Large Language Models (LLMs)
- Natural Language Processing (NLP)
- Machine Learning
- Deep Learning
- Generative AI
- Clinical Reasoning Models

These technologies enable the AI to understand patient conversations and generate medically relevant responses.

7. Working Mechanism

The workflow of AMIE is:

1. Patient enters symptoms.
2. AI understands the natural language input.
3. The model asks follow-up questions.
4. Symptoms are analyzed using medical knowledge.
5. AI predicts possible diseases.
6. Recommendations are generated.
7. Doctor reviews the AI-generated suggestions before making the final diagnosis.

Workflow Diagram:

Patient Symptoms



Natural Language Processing



Large Language Model (AMIE)



Medical Knowledge Analysis



Possible Diagnosis



Recommendations



Doctor Validation

8. Data Sources and Processing

AMIE uses:

- Electronic Health Records (EHR)
- Medical textbooks
- Clinical guidelines
- Research publications
- Patient symptom descriptions
- Medical datasets

Data Processing Steps

- Data collection
- Data cleaning
- Feature extraction
- Model training
- Symptom analysis
- Diagnosis prediction

9. Real-World Implementation

Organizations using AI-assisted diagnosis include:

- Google Research (AMIE)
- Mayo Clinic
- Cleveland Clinic
- NHS (United Kingdom)
- Apollo Hospitals
- Microsoft Healthcare AI
- IBM Watson Health (historically)

Applications include:

- Telemedicine
- Virtual consultations
- Primary healthcare
- Clinical decision support
- Emergency triage

10. Impact and Benefits

Benefits include:

- Faster diagnosis
- Improved accuracy
- Reduced doctor workload
- Better patient experience

- 24/7 availability
- Early disease detection
- Lower healthcare costs
- Increased healthcare accessibility

Studies indicate AI-assisted systems can improve diagnostic support and reduce consultation time in suitable scenarios.

11. Challenges and Limitations

Technical Challenges:

- Requires high-quality medical data
- Bias in training datasets
- Difficulty handling rare diseases

Ethical Challenges:

- Patient privacy
- Data security
- Explainability of AI decisions

Deployment Challenges:

- Integration with hospital systems
- Regulatory approval
- Doctor acceptance
- High implementation cost

AI should assist healthcare professionals, not replace them.

12. Future Scope and Emerging Trends

Future developments include:

- Personalized medicine
- Multimodal AI combining text, images, and lab results
- AI-powered wearable health monitoring
- Predictive disease prevention
- AI-assisted robotic surgery
- Real-time hospital decision support
- Integration with electronic medical records

Generative AI is expected to become an important clinical assistant in the coming years.

13. Conclusion

Google's AMIE represents a significant advancement in AI-assisted healthcare. By combining Large Language Models, Natural Language Processing, and Machine Learning, it can conduct diagnostic conversations and support clinical decision-making.

Although AI cannot replace doctors, it serves as a powerful assistant that improves diagnostic efficiency, accessibility, and patient care. Continued research, ethical safeguards, and clinical validation will further enhance its role in modern healthcare.

14. References

1. Tu T., et al. (2024). Towards Conversational Diagnostic AI. Nature. <https://www.nature.com/articles/s41586-024-07422-z>
2. Google Research. AMIE: Articulate Medical Intelligence Explorer. <https://research.google>
3. World Health Organization (WHO). Ethics and Governance of Artificial Intelligence for Health. <https://www.who.int>
4. National Institutes of Health (NIH). Artificial Intelligence in Healthcare. <https://www.nih.gov>
5. Mayo Clinic. Artificial Intelligence in Medicine. <https://www.mayoclinic.org>
6. IEEE Xplore Digital Library. <https://ieeexplore.ieee.org>
7. PubMed. <https://pubmed.ncbi.nlm.nih.gov>

This topic aligns well with your rubric because it is current, has strong technical depth, includes real-world implementations, and is supported by recent, credible sources.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (59–50%)
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.